Henry Hsu

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EDUCATION

Rice University - Top 20 U.S. University **Aug. 2025 – Dec. 2026 (Expected)**

M.C.S., Computer Science (GPA: 4.00/4.00) *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University **Sep. 2020 – Jun. 2024**

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30) | Class Rank: 6/64

- **CS & Math:** Introduction to Algorithms (A), Operating Systems (A+), Network Programming (A+), Database Systems (A+), Discrete Mathematics (A), Probability (A+), Operations Research I/II (A+)

WORK EXPERIENCE

Software Engineering Intern **Jun. 2026 – Aug. 2026**

Google *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling:** Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Back-End Engineer **Sep. 2024 – Aug. 2025**

MaiAgent Co., Ltd. *Taipei, Taiwan*

- **Open Source:** Contributed **14** merged pull requests to **LlamaIndex** (**15K+** stars), fixing bugs and adding features across its AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, and MCP Client integrations
- **Performance:** Cut response time of 13 high-traffic endpoints by **27.7%** across a **140+** RESTful API surface through SQL query refactoring, connection pooling, and Django caching, eliminating N+1 queries
- **Concurrency & Scale:** Rebuilt knowledge base indexing as an async pipeline on **asyncio** and **Celery** with batch/generator patterns, **3.5x faster** parsing while scaling **3M to 20M+** records through a normalized schema and backward-compatible refactors
- **Correctness:** Took test coverage from a **zero** baseline to **67%** with pytest unit and E2E suites on GitHub Actions, cutting production hotfixes by **90%** initially and **50%** long term

PROJECTS

Relational Database Engine from Scratch | *C++, B+ Tree, Query Optimization, TPC-H* **Aug. 2025 – Dec. 2025**

- **Storage Engine:** Built the C++ persistence layer over a page-level reader/writer API: a buffer manager with **LRU** eviction and page pinning, slotted-page record storage, and a **B+ tree** index supporting range scans
- **Query Execution:** Implemented the relational operators, selection, aggregation with group-by, scan join, and a **sort-merge join** backed by external merge sort that spills runs to disk when inputs exceed the buffer pool
- **SQL Front End & Optimizer:** Compiled SQL into logical plans, ordered joins by cost from cardinality statistics, and chose scan vs. sort-merge join by buffer-pool fit; ran the **12-query TPC-H** suite at scale factor 1 (**1 GB**)

Chat Bar - Real-Time Multiplayer Game Server | *C/C++ (Socket Programming, SFML), MySQL, SHA-256* **Oct. 2023 – Jan. 2024**

- **Server-Client Architecture:** Designed and implemented a real-time multiplayer game server using **TCP socket programming** with multi-threaded client handling, supporting concurrent connections for chat and gameplay state synchronization
- **User Management & Client:** Built authentication with SHA-256 password hashing and MySQL persistence plus ranking features backed by optimized queries; developed the SFML client with an event-driven loop for movement, group chat, and live state updates

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Taiwan national competition; built an urban noise analysis agent

AI Workshop Outstanding Award (Top 3 of 50 teams) - Multi-agent tutoring system, NYCU Computer Science

SKILLS

Languages: C/C++, Python, Java, TypeScript, SQL | **Systems:** Socket programming, multithreading, asyncio, Celery, Linux, Docker
Data: PostgreSQL, MySQL, MongoDB, Elasticsearch, Neo4j | **Testing:** pytest, JUnit, Vitest, Jasmine | **Tooling:** git, GitHub Actions, Nginx, Shell | **Web:** Django, FastAPI, React
Quantitative: numpy, pandas, scikit-learn, PyTorch | Genetic Algorithms, Bayesian Optimization, Operations Research